Scenario 5: High Application Latency Using Datadog

Objective

Use Datadog APM to identify a slow workload before investigating Kubernetes and applying a focused recovery.

Trigger the Incident

Run this from the repository root against your own lab environment:

```
./scripts/chaos/inject-latency.sh ecommerce 3000
```

The request reaches one backend pod because chaos state is stored per process. Generate application traffic and wait for APM ingestion before troubleshooting. Repeat through the documented port-forward method only when you want every backend pod affected.

Situation

Users report that ecommerce pages load slowly. Do not restart pods before scoping the problem.

Symptoms

- Ecommerce responses are intermittently slow.
- Pods remain Ready.

What You Know

- Datadog reports elevated p95 latency.
- The issue affects `ecommerce-backend` while pods remain Ready.
- The start time does not match an AWS or database change.

Start With Datadog

Open the ecommerce dashboard and **SRE Lab Scenario Signals**. Inspect `trace.express.request` p95, throughput, errors, CPU, memory, replicas, logs, and the **High p95 latency** monitor. In APM, open slow `ecommerce-backend` traces and note which spans consume the time.

Troubleshooting

Use the Datadog service and time window to scope Kubernetes checks.

```
kubectl get pods -n ecommerce
kubectl top pods -n ecommerce
kubectl get deployment -n ecommerce
kubectl describe pod <pod-name> -n ecommerce
kubectl logs <pod-name> -n ecommerce
kubectl get events -n ecommerce --sort-by=.lastTimestamp
```

Compare slow traces with CPU, memory, replica count, events, and logs. Inspect workload behavior instead of changing unrelated infrastructure.

Important Clues

Determine why one or more backend containers add time before every response. Use trace duration and the chaos-state endpoint as evidence.

```
curl -s https://ecommerce.$(cat .lab-domain)/api/chaos
```

Root Cause

Record the workload behavior that accounts for the trace duration while Kubernetes resource metrics remain normal. Confirm it with the instructor.

Fix

Clear the workload's injected latency. Chaos state is per process, so repeat until every backend pod is reset, or reset each pod through port-forwarding as documented in the README.

```
./scripts/chaos/reset.sh ecommerce
```

Validation

Generate normal requests. Confirm p95 latency and traces return to baseline, pods remain healthy, Kubernetes metrics remain normal, and the Datadog monitor recovers.

DevOps Lesson

Use traces to isolate the slow workload before taking action. A restart may hide evidence without explaining the cause.

STAR Method

Situation

Summarize the latency report and initial scope.

Task

Explain the need to identify the affected workload before acting.

Action

Describe the dashboard, trace, Kubernetes, log, and workload-state evidence.

Result

State how normal latency and healthy pods were verified.

Natural Spoken Version

Use the matching example in DevOps STAR Scenarios after completing the lab.